

One Week Online Workshop

THE PHOTONICS SPECTRUM 2026



Oct 01- 07, 2026

10:00AM - 4:00PM

Sponsored by

OPTICA

Jointly Organized by



सह वीर्य करवावहे
IETE Delhi Centre



ABOUT OPTICA

OPTICA

Optica (formerly OSA), Advancing Optics and Photonics Worldwide, is the society dedicated to promoting the generation, application, archiving and dissemination of knowledge. Founded in 1916, it is the leading organization for scientists, engineers, business professionals, students and others interested in the science of light. Optica's renowned publications, meetings, online resources and in-person activities fuel discoveries, shape real-life applications and accelerate scientific, technical and educational achievement.



ABOUT IETE

The Institution of Electronics and Telecommunication Engineers (IETE) is India's leading recognised professional society devoted to the advancement of Science and Technology of Electronics, Telecommunication & IT. Founded in 1953. The IETE is the National Apex Professional body of Electronics and Telecommunication, Computer Science and IT Professionals. It serves more than 1,25,000 members (including Corporate, Student and ISF members) through various 63 Centres, spread all over India and abroad. The Institution provides leadership in Scientific and Technical areas of direct importance to the national development and economy. The IETE focuses on advancement of the Science and Technology of Electronics, Telecommunication, Computers, Information Technology and related areas. Towards this end the Institution promotes and conducts basic engineering and continuing technical education programmes for human resource development. IETE has been engaged in bringing out publications ever since its inception in 1953. At present IETE publishes journals, books, conference proceedings etc targeted at specific readership.

ABOUT THE PHOTONICS SPECTRUM 2026

The Photonics Spectrum 2026 is a seven-day workshop focused on emerging developments in Optical Interactions, Integrated Photonics, Materials, devices and Quantum Communication.

The workshop brings together technical lectures, hands-on training, industry-oriented learning, professional development sessions, and opportunities for interaction between academia and industry. The programme will feature experts from academia and industry, providing participants with exposure to contemporary research, technological developments, practical skills, and career-oriented perspectives.

The workshop will be conducted in ONLINE mode. Each day will comprise approximately 5 hours of activities, including: 3 hours of technical lectures, 2 hours of hands-on activities.

Whom to apply?

This workshop is a must-attend for professionals, researchers, and enthusiasts such as Faculty Members, Undergraduate students, Postgraduate students, Research scholars and Industry Professionals, seeking to stay on top of the rapidly evolving world of intelligent optical networks, nano-materials, integrated optics, and optical sensors. Especially in this workshop, Max of 500 participants from the worldwide education institutions can join for proficient training and learning. So, don't miss this opportunity to expand your understanding and explore the exciting possibilities of next-generation communication technologies.

How to apply?

See Details at:

<https://vidyasagar777.github.io/The-Photonics-Spectrum-2026/>

<https://www.opticansut.in/events>

Register NOW



Participants are encouraged to

Apply through

<https://forms.gle/RvtqaH3o34Dxz22N8>

For further assistance join what's app group

<https://chat.whatsapp.com/KgnHOi9mOYd6Eu8y1YPXwy>

SCHEDULE

“Certificate of Participation” will be provided to all registered participants having attendance $\geq 80\%$.

Time Slot (IST) (UTC+05:30)	Day 1 01 Oct 2026	Day 2 02 Oct 2026	Day 3 03 Oct 2026	Day 4 04 Oct 2026	Day 5 05 Oct 2026	Day 6 06 Oct 2026	Day 7 07 Oct 2026
10:00 AM – 11:00 AM	Inauguration	Session 5	Session 10	Session 15	Session 20	Session 25	Session 30
11:00 AM – 12:00 PM	Session 1	Session 6	Session 11	Session 16	Session 21	Session 26	Session 31
12:00 PM – 01:00 PM	Session 2	Session 7	Session 12	Session 17	Session 22	Session 27	Session 32
Break							
2:00 PM – 3:00 PM	Session 3	Session 8	Session 13	Session 18	Session 23	Session 28	Valedictory
3:00 PM – 4:00 PM	Session 4	Session 9	Session 14	Session 19	Session 24	Session 29	

* All Sessions will be in online mode.

SPEAKERS

Day 1 - 01 Oct 2026



“AI in Quantum Key Distribution (QKD)”

[session - 1]

Dr. Avinash Upadhyay

Research Associate, TTDF Project 604, NSUT Delhi
Delhi, India

TTDF Project: Quantum Communication for Telecommunication Optical Network Security



“2D/3D heterostructures”

[session - 2]

Dr. Neeraj Goel

Assistant Professor, Netaji Subhas University of Technology
Delhi, India



“Basics of Quantum Communications”

[session - 3]

Dr. Vivek Narayan

Retd. Deputy Director General
Department of Telecommunications, Govt. of India
New Delhi, India

2022 Optica Ambassador



**“Multiresonant Plasmonic Metasurfaces for Biosensing:
Principles, Technologies, and Applications”**

[session - 4]

Dr. Jhonattan Córdoba Ramírez

Assistant Professor, Universidade Federal de Minas Gerais (UFMG)
Belo Horizonte, Brazil

SPEAKERS

Day 2 - 02 Oct 2026



“QKD Protocols, PQC algorithms, and status of development of Quantum technology and Standardization in India”

[session - 5]

Dr. Vivek Narayan

Retd. Deputy Director General
Department of Telecommunications, Govt. of India
New Delhi, India

Social Media Officer - Optica TG



“From Light to Sensing: Fundamentals and Engineering of SPR Sensors”

[session - 6]

Dr. Nitesh Mudgal

Associate Professor, Poornima College of Engineering, Jaipur
Jaipur, India
Social Media Officer of Optica TG - Integrated Photonics (PI)



“Intelligent Photonic and Semiconductor Devices for Next-Generation Sensing and Healthcare”

[session - 7]

Dr. Deepak Punetha

Assistant Professor, NIT Allahabad
Prayagraj, India



“Emerging Group IV Semiconductor Photodetectors for Next-Generation Photonic Integrated Circuits”

[session - 8]

Dr. Syed Sadique Anwer Askari

Assistant Professor, Aligarh Muslim University
Aligarh, India

Optica Ambassador



“Principles of Effective Communication: Speaking to Broad Audiences”

[session - 9]

Dr. Travis Sawyer

Assistant Professor, University of Arizona
Tucson, USA

SPEAKERS

Day 3 - 03 Oct 2026

“From Fiber to Brain Implants: Navigating a Career in Bio-photonics & Neurotechnology”

[session - 10]

Dr. Garima Bawa

Postdoctoral Research Associate, CREOL, University of Central Florida
Orlando, USA

“Speckle Interference for Controlled Synthesis and Characterization of Spatial Statistics of Laser Speckles”

[session - 11]

Dr. Abhijit Roy

Assistant Professor, Manipal Academy of Higher Education (MAHE)
Manipal, India

Chair of Optica TG - IH Holography and Diffractive Optics

“Large Area Manufacturing of Micro/Nano Optics”

[session - 12]

Dr. Vinod Mishra

Senior Technical Officer and Asst. Professor, CSIR and AcSIR
Chandigarh, India

Chair of Optica TG - FM Optical Fabrication and Testing

“Two-photon polymerized nanostructures as high-sensitivity homogeneous and reproducible SERS substrates”

[session - 13]

Dr. Tatevik Chalyan

Vrije Universiteit Brussel
Brussels, Belgium

“Detection of nanoplastics through the use of surface enhanced Raman spectroscopy ”

[session - 14]

Dr. Aisha Bibi

Photonics Integration Engineer, Saliency Labs Ltd
London, United Kingdom

SPEAKERS

Day 4 - 04 Oct 2026

Chair - Optica TG



“Computational Photonics: Computational Tools in Design and Fabrication of Photonic Devices”

[session - 15]

Dr. Saswatee Banerjee

Senior Optical Scientist, Photonic Wave Solutions LLC
San Jose, USA

Chair of Optica TG - FA Lighting Applications

Industry Professional



“Materials Selection for Optical and Photonic Applications Using Ansys Granta EduPack”

[session - 16]

Dr. Rohit Sangwan

Technical Lead, ARK Infosolution Pvt. Ltd
New Delhi, India



“2D Materials-Based Photodetectors: From Material Engineering to Advanced Device Applications”

[session - 17]

Dr. Rahul Kumar

Assistant Professor, Institute of Infrastructure Technology
Research and Management (IITRAM)
Ahmedabad, India



“Vitals of Medical Data: Foundations and Applications”

[session - 18]

Dr. Amit Singhal

Associate Professor, NSUT
Delhi, India

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“Near-Eye Displays and Human Vision”

[session - 19]

Dr. Tatjana Pladere

Associate Professor, University of Latvia
Riga, Latvia

SPEAKERS

Day 5 - 05 Oct 2026



“Smart Optical Materials: From Molecular Design to Light-Responsive Engineering”

[session - 20]

Dr. Swapnil L. Sonawane

Assistant Professor, NSUT Delhi
Delhi, India

2026 Optica Ambassador



“The Future Is Integrated: From Photonic Integrated Circuits to Compact OCT Systems”

[session - 21]

Dr. Fabian Ruf

Integrated Optics & Photonics Designer, Carl Zeiss AG
Oberkochen, Germany

Industry Professional



“From Components to Systems: Introduction to Photonics Simulation with Ansys Lumerical”

[session - 22]

Dr. Jayakrishnan K

Application Engineer - Optics, ARK Infosolutions
Chennai, India

Industry Professional



“Rapid Prototyping in Electronics: In-House PCB Milling for Modern Engineering Education”

[session - 23]

Arun Kumar

Director, Ambition Technologies
Delhi, India

2026 Optica Ambassador



“Multi-dimensional Frequency-Bin Entanglement-Based Quantum Key Distribution Networks”

[session - 24]

Dr. Juan Rafael Álvarez

Optica Ambassador 2026, Telecom Paris
Palaiseau, France

SPEAKERS

Day 6 - 06 Oct 2026



“Semiconductor Optoelectronic Devices for Photonics”

[session - 25]

Dr. Balwinder Raj

Associate Professor, NIT Jalandhar - Punjab
Jalandhar, India



“Device -Circuit Mixed mode TCAD Simulation”

[session - 26]

Dr. Ashish Maurya

Assistant Professor, Bennett University
Greater Noida, India



“Building Your Scientific Community: Mentors, Sponsors, Peers, and Networks”

[session - 27]

Dr. Saheer Cheemadan

Optica Ambassador (2026), SOHSS Areekode, Kerala, India
Kozhikode, India



“Demonstration of Underwater sensor Communication challenges & Indian Market”

[session - 28]

Dr. Anirudha Kulkarni

Director, RFAQUA SOLUTIONS PUNE
Pune, India



Photonic Crystal Waveguides: Design Strategies, and Emerging Applications

[session - 29]

Dr. Chandra Prakash

Assistant Professor, National Institute of Technology Kurukshetra
Kurukshetra, India

Social Media Officer of Optica TG - Integrated Photonics (PI)

Chair - Optica TG



“Computational Imaging: Designing Optics for What We Want to Know”

[session - 30]

Dr. John J. Healy

Assistant Professor, University College Dublin
Dublin, Ireland

Chair of Optica TG - IR-IAPD Image Sensing and Pattern Recognition

Chair - Optica TG



“Leveraging Optical Polarization for Biomedical Imaging”

[session - 31]

Dr. Mahima Sharma

Postdoctoral Associate, IIT Hyderabad
Hyderabad, India

Chair of Optica TG - BS Tissue Imaging and Spectroscopy

2026 Optica Ambassador



“The Future of UK Photonics: Key Takeaways from the New Photonics Roadmap”

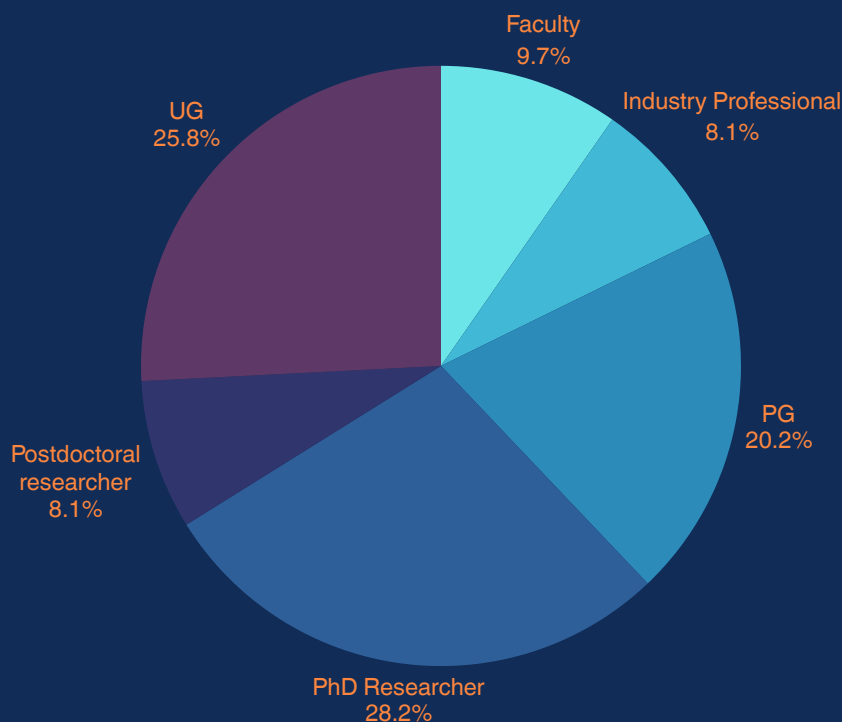
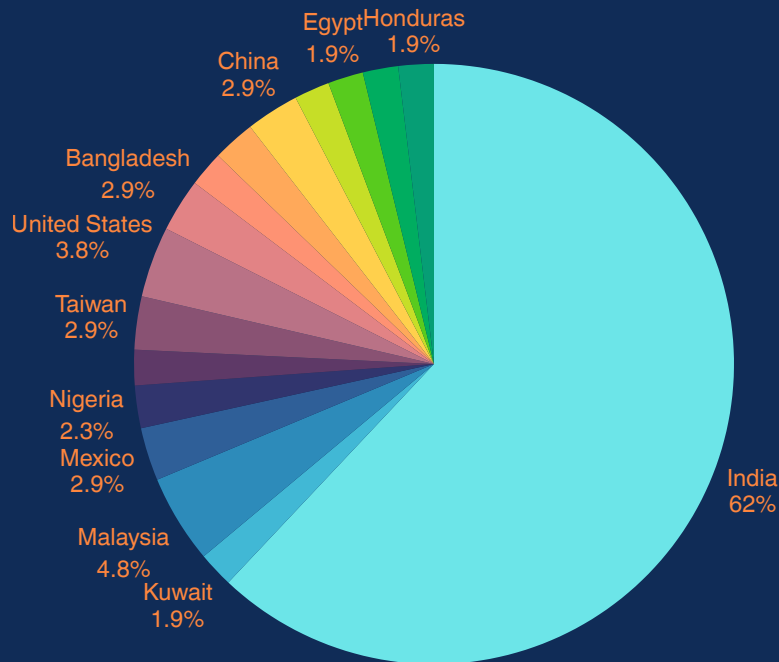
[session - 32]

Dr. Lewis Hill

General Manager, UK Photonics Leadership Group
Manchester, United Kingdom

2026 Participants

The Photonics Spectrum 2026 is expected to bring together around 550 attendees from ~ 100 universities and organizations across the world. The diverse participation from institutions spanning multiple countries reflects the workshop's growing international reach and its role in fostering collaboration and exchange within the global scientific community.



ORGANIZING COMMITTEE

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»»»»



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**Vice Chancellor,
NSUT Delhi**

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»»»»



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NSUT Delhi**



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NSUT Delhi**



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Co-Advisor - NSUT
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COORDINATORS**

««««

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NSUT Delhi



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Rawat**

Professor - ECE
NSUT Delhi



**Dr. Urvashi
Bansal**

Asst. Professor - ECE
NSUT Delhi



**Dr. Shweta
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Asst. Professor - ECE
NSUT Delhi



Dr. Neeraj Goel

Asst. Professor - ECE
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Dr. Ankit Garg

Asst. Professor - ECE
NSUT Delhi

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